

Exam 1 Review: Proof Templates and Practice

Chapter 0

Proof Template – Indirect Proofs

When using an indirect proof to prove a statement: “If $[P]$, then $[Q]$ ”

- ❖ It is a **good idea** to start a proof by contrapositive with
 - “We prove the contrapositive: If $[\text{not } Q]$, then $[\text{not } P]$. Assume $[\text{not } Q]$...”

- ❖ It is a **good idea** to start a proof by contradiction with either
 - “We argue by contradiction. Suppose $[P]$ and $[\text{not } Q]$ ”
 - OR
 - “Suppose for a contradiction that $[P]$ and $[\text{not } Q]$.”

Practice – Prove Using Contrapositive

Theorem. For every integer n , if n^3 is odd, then n is odd.

pf. We prove the contrapositive: for every integer n , if n is even then n^3 is even. Since n is even, there exists $k \in \mathbb{Z}$ such that $n = 2k$. So, $n^3 = (2k)^3 = 8k^3 = 2(4k^3)$. Since the integers are closed under multiplication, $4k^3 = m \in \mathbb{Z}$ and $n^3 = 2m$ is an even integer.

pf. by contradiction. Assume, by way of contradiction that n^3 is odd and n is even. Then there exists $k \in \mathbb{Z}$ such that $n = 2k$ and thus $n^3 = (2k)^3 = 8k^3 = 2(4k^3)$. Since the integers are closed under multiplication, $4k^3 = m \in \mathbb{Z}$ and $n^3 = 2m$ is an even integer. Contradiction.

Bonus Activity: Prove using contradiction.

Proof Template for Sets

1. To show that $T \subseteq S$:
 - I. “Fix $t \in T$.”
 - II. Show that $t \in S$.
 - III. “Since $t \in T$ was arbitrary, $T \subseteq S$.”
2. To show that $T = S$:
 - Show that $T \subseteq S$.
 - Show that $S \subseteq T$.

Practice

Activity: Let $A = \{4k + 1 : k \in \mathbb{Z}\}$ and $B = \{4k - 3 : k \in \mathbb{Z}\}$. Prove that $A = B$.

pf. $(A \subseteq B)$ let $a \in A$. Then $a = 4k + 1$ for some $k \in \mathbb{Z}$. Then

$$a = 4k + 1 + 4 - 4 = 4(k+1) - 4 + 1 = 4(k+1) - 3. \quad \text{Since } k+1 \in \mathbb{Z}$$

$a \in B$. Since $a \in A$ was arbitrary, $A \subseteq B$.

$(B \subseteq A)$ let $b \in B$. Then $b = 4k - 3$ for some $k \in \mathbb{Z}$. Then

$$b = 4k - 3 + 4 - 4 = 4(k-1) + 1. \quad \text{Since } k-1 \in \mathbb{Z}, \quad b \in A.$$

Since $b \in B$ was arbitrary, $B \subseteq A$.

Chapter 1

Mathematical Induction

Basic Template:

- “We argue by induction.”
- “Base Case: If $n = 1$, then...”
 - Prove true for $n = 1$.
- “Induction Step: Assume [the statement] is true for some $n \in \mathbb{N}$.”
 - Prove true for $n + 1$.

Practice – Prove Using Induction

Theorem. If $n \in \mathbb{N}$, then $\sum_{i=1}^n \frac{1}{i(i+1)} = 1 - \frac{1}{n+1}$.

pf. We argue by induction.

Base Case: If $n=1$ then $\sum_{i=1}^1 \frac{1}{i(i+1)} = \frac{1}{1(1+1)} = \frac{1}{2}$ and $1 - \frac{1}{1+1} = \frac{1}{2}$ so holds when $n=1$.

Inductive Step: Assume $\sum_{i=1}^n \frac{1}{i(i+1)} = 1 - \frac{1}{n+1}$ for some $n \in \mathbb{N}$. Then,

$$\begin{aligned} \sum_{i=1}^{n+1} \frac{1}{i(i+1)} &= \sum_{i=1}^n \frac{1}{i(i+1)} + \frac{1}{(n+1)(n+1+1)} = 1 - \frac{1}{n+1} + \frac{1}{(n+1)(n+2)} = 1 - \frac{1}{n+1} \left(1 - \frac{1}{n+2}\right) \\ &= 1 - \frac{1}{n+1} \left(\frac{n+2-1}{n+2}\right) = 1 - \frac{1}{n+2} = 1 - \frac{1}{(n+1)+1}. \end{aligned}$$

So true for $n+1$.

Bonus Activity: Prove in a different way. (Hint: Break $\frac{1}{i(i+1)}$ into a difference)

Note that $\frac{1}{i(i+1)} = \frac{1}{i} - \frac{1}{i+1}$ so that $\sum_{i=1}^n \frac{1}{i(i+1)} = \sum_{i=1}^n \left(\frac{1}{i} - \frac{1}{i+1}\right) = 1 - \frac{1}{2} + \frac{1}{2} - \frac{1}{3} + \dots$

$$\frac{1}{n} - \frac{1}{n+1} = 1 - \frac{1}{n+1}.$$

Absolute Value & Distance

Definition. The **absolute value function** $|\cdot| : \mathbb{R} \rightarrow \mathbb{R}$ is defined by

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

Recall: $|a - b|$ is the distance between $a, b \in \mathbb{R}$.

Triangle Inequality: $|a + b| \leq |a| + |b|$.

Proof Technique: Argue by cases or use the triangle inequality .

Rational/Irrational Numbers

The rational numbers $\mathbb{Q}$ are the numbers of the form $\frac{p}{q}$ where $p, q \in \mathbb{Z}$ and $q \neq 0$.

A real number is **irrational** if it is not rational.

Proof Technique: To show that a number is irrational, usually use a proof by contradiction.

Practice – Prove the Theorem

Theorem. If $x \geq 0$ is irrational, then $\sqrt{x}$ is irrational.

Pf. Assume, by way of contradiction that $x \geq 0$ is irrational and $\sqrt{x}$ is rational.

Then there exist $p, q \in \mathbb{Z}$, $q \neq 0$, such that $\sqrt{x} = \frac{p}{q}$. Thus

$$x = (\sqrt{x})^2 = \left(\frac{p}{q}\right)^2 = \frac{p^2}{q^2}. \quad \text{Both } p^2, q^2 \in \mathbb{Z} \text{ and } q^2 \neq 0$$

Therefore x is rational. Contradiction.

Proof Template for sup/inf

1. To show that y is an upper bound for $S \subseteq \mathbb{R}$:
 - I. “Fix $s \in S$.”
 - II. Show that $s \leq y$.
 - III. “Since $s \in S$ was arbitrary, y is an upper bound for S .”
2. To show that $y = \sup S$:
 - Show that y is an upper bound for S . Hence $\sup S \leq y$.
 - Show that if u is an upper bound for S , then $y \leq u$. Hence y is the least upper bound for S .

Activity: Write a version of the above for lower bounds and the infimum

Practice – Prove the Theorem

Theorem. Let $S \subseteq \mathbb{R}$ be a non-empty bounded set and let

$$-S = \{-x : x \in S\}.$$

Then:

1. y is an upper bound for S if and only if $-y$ is a lower bound for $-S$.

2. $\inf(-S) = -\sup S$.

Pf. of 1. ($\Rightarrow$) let y be an upper bound for S . let $x \in -S$. Then $-x \in S$ and $y \geq -x \Rightarrow -y \leq x$. Since $x \in -S$ was arbitrary, $-y$ is a lower bound for $-S$.

($\Leftarrow$) let $-y$ be a lower bound for $-S$. let $x \in S$. Then $-x \in -S$ and $-y \leq -x \Rightarrow y \geq x$. Since $x \in S$ was arbitrary, y is an upper bound for S .

Pf of 2. Since $\sup S$ is an upper bound for S , by 1., $-\sup S$ is a lower bound for $-S$. let z be another lower bound of $-S$. By 1., $-z$ is an upper bound for S . so $-z \geq \sup S \Rightarrow z \leq -\sup S$. Since z was an arbitrary lower bound, $-\sup S$ is greatest lower bound, that is $\inf(-S) = -\sup S$.

Chapter 2

Proof Template for explicit limits

Given an explicit sequence, like $s_n = \frac{1}{n}$, to show that $\lim s_n = s$ using only the definition:

- I. “Fix $\epsilon > 0$.”
- II. “Let $N = (\textit{something})$.” [this requires scratch work]
- III. “Assume $n > N$ ”
- IV. Show that $|s_n - s| < \epsilon$.
- V. “Since $\epsilon > 0$ was arbitrary, $\lim s_n = s$.”

Scratch Work:

$$\left| \sqrt{n^3-1} - n^{3/2} \right| = \left| \sqrt{n^3-1} - n^{3/2} \cdot \frac{\sqrt{n^3-1} + n^{3/2}}{\sqrt{n^3-1} + n^{3/2}} \right| =$$

$$\left| \frac{n^3-1 - n^3}{\sqrt{n^3-1} + n^{3/2}} \right| = \frac{1}{\sqrt{n^3-1} + n^{3/2}} < \frac{1}{n^{3/2}}$$

Practice

Activity: Prove, using only the definition of the limit, that

$$\lim \sqrt{n^3-1} - n^{3/2} = 0.$$

PF. Let $\varepsilon > 0$. Set $N = \frac{1}{\varepsilon^{2/3}}$. Assume $n > N$. Then

$$\begin{aligned} \left| \sqrt{n^3-1} - n^{3/2} - 0 \right| &= \left| (\sqrt{n^3-1} - n^{3/2}) \frac{\sqrt{n^3-1} + n^{3/2}}{\sqrt{n^3-1} + n^{3/2}} \right| = \left| \frac{n^3-1 - n^3}{\sqrt{n^3-1} + n^{3/2}} \right| \\ &= \frac{1}{\sqrt{n^3-1} + n^{3/2}} < \frac{1}{n^{3/2}} < \frac{1}{N^{3/2}} = \frac{1}{\left(\frac{1}{\varepsilon^{2/3}}\right)^{3/2}} = \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, $\lim \sqrt{n^3-1} - n^{3/2} = 0$.

Common Tricks

If your assumptions include the fact that $\lim s_n = s$, your proof will probably have a sentence of the form

- Since $\lim s_n = s$, there exists N such that: if $n > N$, then $|s_n - s| < \square$

If your assumptions include the fact that $\lim s_n = s$ and $\lim t_n = t$, your proof will probably have sentences of the form

- Since $\lim s_n = s$, there exists N_1 such that: if $n > N_1$, then $|s_n - s| < \square$
- Since $\lim t_n = t$, there exists N_2 such that: if $n > N_2$, then $|t_n - t| < \square$
- Let $N = \max(N_1, N_2)$

scratch work: $|2s_n + 3t_n - (2s + 3t)| \leq 2|s_n - s| + 3|t_n - t| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2}$.

Practice – Use only the definition of the limit

Theorem. If $\lim s_n = s$ and $\lim t_n = t$, then

$$\lim (2s_n + 3t_n) = 2s + 3t.$$

pf. Let $\varepsilon > 0$. Since $\lim s_n = s$, there exists some N_1 such that if $n > N_1$, $|s_n - s| < \frac{\varepsilon}{4}$. Since $\lim t_n = t$, there exists some N_2 such that if $n > N_2$, $|t_n - t| < \frac{\varepsilon}{6}$. Set $N = \max\{N_1, N_2\}$. Assume $n > N$. Then

$$\begin{aligned} |(2s_n + 3t_n) - (2s + 3t)| &= |2(s_n - s) + 3(t_n - t)| \leq \\ &2|s_n - s| + 3|t_n - t| < 2\left(\frac{\varepsilon}{4}\right) + 3\left(\frac{\varepsilon}{6}\right) = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

Since $\varepsilon > 0$ was arbitrary, $\lim 2s_n + 3t_n = 2s + 3t$.